8.5 Using Recursive Rules with Sequences

Name _____ Period ____ Date _____

Algebra II 8.5 · Student sheet

I can…

- find the terms of a sequence from a recursive rule.
- write recursive rules for sequences.
- translate between explicit and recursive rules.

Vocabulary · fill in the missing word

Explicit rule — gives a_n as a _____ of n

Recursive rule — gives the first _____(s) and an equation for a_n using earlier terms

Arithmetic — a_1 = first _____, $a_n = a_{n-1} + d$

Geometric — a_1 = first _____, $a_n = r \cdot a_{n-1}$

Part 1 · Fish pond

A pond has 400 fish. Each year, 20% of the fish are caught, and then 100 new fish are added.

Year	1	2	3	4	5	6
Fish	400					

Recursive rule: a_1 = _____, a_n = _____ Where are the numbers heading? _____

Part 2 · Notes and examples (fill in each note, then work the example)

Recursive rule: A recursive rule gives the _____ and how to get each term from the _____.

Example 1

Write the first six terms of each sequence.

a. $a_1 = 2,\ a_n = a_{n-1} + 5$ b. $f(0) = 1,\ f(1) = 1,\ f(n) = f(n-1) + f(n-2)$

Recursive rules: Arithmetic: a_n = _____ Geometric: a_n = _____

Example 2

Write a recursive rule for each sequence.

a. 3, 13, 23, 33, … b. 16, 40, 100, 250, …

Example 3

Write a recursive rule for each sequence.

a. $a_n = -6 + 8n$ b. $a_n = -3(2)^{n-1}$

Example 4

Write an explicit rule for each sequence.

a. $a_1 = -5,\ a_n = a_{n-1} - 2$ b. $a_1 = 10,\ a_n = (\frac{1}{2})a_{n-1}$

Part 3 · Partner practice

Partner 1

Write a recursive rule for 1, 4, 16, 64, …

Partner 2

Write an explicit rule for $a_1 = 12,\ a_n = a_{n-1} + 7$.